MIL-M-24519D(NAVY)
AMENDMENT 8
7 February 1986
SUPERSEDING
AMENDMENT 7
25 June 1985

MILITARY SPECIFICATION

MOLDING PLASTICS, ELECTRICAL, THERMOPLASTIC

This amendment forms a part of Military Specification MIL-M-24519D(NAVY), dated 15 August 1983, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Delete and substitute: "Molding Plastics, Electrical, Thermoplastic"

1.2: Under "PT-F" in Types and Description columns, add "GPT-7.5 Thermoplastic Polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 7.5 glass reinforced, flame resistant".

PAGE 2

* 1.2: Under "GST-40F" in Types and Description columns, add "MLT-20F Mineral plastic polytherimide compound, 20 percent mineral filled, flame resistant".

PAGE 3

3.2.2: Heading title: Delete "MIL-STD-595" and substitute "FED-STD-595".

3.2.2: Add under "PT-F" in types, FED-STD-595, and Chip number columns "GPT-7.5", "white", and "37778".

PAGE 4

3.2.2: Under Chip number column: Delete "10032" and substitute "10G32". Also under Types, FED-STD-595, and Chip number column, add "GST-40F", "Black", and "17038".

* 3.2.2: Add under "GST-40F" in types, FED-STD-595, and Chip number columns "MLT-20F", "Greenish/Brown", and "24098/24201 2/".

* 4.3.1: Delete "TABLES I and III" and substitute "TABLE I".

* 4.4.3: Delete "TABLES II and III" and substitute "TABLE III".

* At bottom of page, add new footnote: "2/ The greenish/brown chip number falls between green chip 24098 and brown chip 24201".

PAGES 7 and 8

* TABLE II: Delete and substitute the table attached.

PAGE 9

* TABLE III, Lower portion of Table: Delete and substitute the Table attached.

PAGE 11

* 4.5.2.5b: After "GST-40F" add "MLT-20F".

AMSC N/A FSC 5970

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MIL-M-24519B(NAVY)
AMENDMENT 8

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

User Activities:
NAVY - SH, AS, MC

Preparing activity:
NAVY - EC
(Project No. 5970-N645(EC))

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TABLE II. Property values for qualification tests of thermoplastic molding compounds.

Property to be tested	Value required for each type of compound													Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	
Arc resistance	50	55	65	60	70	80	70	70	85	85	85	30	85	Sec
Compressive strength, 10 percent deflection	12,000	12,500	13,500	14,000	16,200	13,500	17,500	16,200	19,000	20,000	22,000	20,000	20,000	lb/in^2
Dielectric constant:														
At 100 Hz	3.6	3.8	3.9	3.9	4.1	3.8	4.1	4.1	3.4	3.4	3.6	4.3	3.5	---
At 10^6 Hz	3.4	3.6	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.6	4.2	3.5	---
Dissipation factor:														
100 Hz	0.005	0.0050	0.005	0.005	0.010	0.003	0.010	0.005	0.007	0.012	0.004	0.002	0.004	---
10^6 Hz	0.03	0.0300	0.03	0.03	0.03	0.02	0.03	0.03	0.009	0.012	0.009	0.007	0.009	---
Dielectric strength:														
Short-time test	350	360	370	375	400	400	400	400	450	450	450	320	450	Volt/mil
Short-time test	325	330	340	350	375	375	375	375	430	430	430	300	430	
Dielectric breakdown:														
Short-time test	43	45	46	46	50	50	50	50	56	56	56	40	56	Kilovolt
Short-time test	40	41	42	42	46	46	46	46	54	54	54	37	54	
Flame resistance:														
Ignition time	50	55	55	50	55	55	50	55	65	65	65	50	65	Sec
Burning time	35	50	50	50	20	50	20	20	20	20	20	20	20	
Flexural strength	13,000	15,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	22,500	lb/in^2
Flammability	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	---
Heat-deflection temperature	160	270	360	375	385	300	400	385	400	400	400	475	410	oF
Impact strength	0.6	0.6	0.7	0.8	1.1	0.8	1.1	1.1	0.9	1.0	1.1	1.1	0.6	ft-lb/in

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Property to be tested	Value required for each type of compound													Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	
Arc resistance	50	55	65	70	70	80	70	70	85	85	85	30	85	Sec
Dielectric constant at 1 megahertz	3.4	3.6	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.6	4.0	3.5	
Dissipation factor at 1 megahertz	0.03	0.03	0.03	0.03	0.03	0.02	0.03	0.03	0.009	0.012	0.009	0.007	0.009	
Dielectric strength short time	350	360	370	375	400	400	400	400	450	450	450	320	450	Volts/mil
Flexural strength	13,000	15,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	22,500	lb/in^2
Impact strength, side	0.60	0.60	0.70	0.8	1.10	1.0	1.1	1.1	0.9	1.0	1.1	1.1	0.6	ft-lb/in
Water absorption	0.40	0.35	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.08	0.60	Percent

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TABLE II. Property values for qualification tests of thermoplastic molding compounds. - continued

Property to be tested	Value required for each type of compound													Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	
Tensile strength	8,000	8,400	11,500	12,000	15,000	12,000	15,000	15,000	16,000	18,700	23,000	16,000	13,000	lb/in^2
Water absorption	0.40	0.35	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.08	0.60	Percent
Heat resistance	50	50	50	50	50	70	50	50	85	85	85	95	95	Percent
Dimensional stability (high temperature)	0.2	0.2	0.2	0.2	0.2	.12	0.2	0.2	0.1	0.1	0.1	0.1	0.1	Percent
Volume resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1x10^{11}$	$1.0x10^9$	$1.0x10^9$	$1x10^{11}$	$1x10^{11}$	$1x10^{10}$	$1.0x10^{12}$	$1x10^{11}$	Ohm
Surface resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$3.5x10^{10}$	$1.0x10^9$	$1.0x10^9$	$1x10^{10}$	$1x10^{10}$	$1x10^{10}$	$1.0x10^{11}$	$1x10^{10}$	Ohm
Tracking index	180	180	180	180	180	125	180	180	135	135	135	150	140	Volt
Toxicity when heated:														
Carbon dioxide	15,000	15,000	15,000	15,000	15,000	10,000	15,000	15,000	5,000	5,000	5,000	1,500	5,000	
Carbon Monoxide	1,000	1,000	1,000	1,000	1,000	500	1,000	1,000	250	250	250	200	250	
Ammonia	2,500	2,500	2,500	2,500	2,500	50	2,500	2,500	20	20	20	100	20	
Aldehydes as H.CHO	50	50	50	50	50	50	50	50	20	20	20	20	20	p/m
Cyanide as HCN	60	60	60	60	60	50	60	60	20	20	20	0	20	
Oxides of nitrogen as NO_2	100	100	100	100	100	60	100	100	50	50	50	25	50	
Hydrogen chloride	100	100	100	100	100	50	100	100	50	50	50	0	50	

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